

A physics study group builds a small model of a swing. A point mass M hangs from a fixed pivot on a light string of length L . Let θ be the angle the swing makes with the downward vertical.

At $t = 0$ the swing is at $\theta = 0$, but the mass has a velocity of magnitude v_0 directed normal to the string. At some later time t' , the tension in the string returns to zero.

Assumptions

1. The string is massless and inextensible, of length L .
2. The pivot is fixed; the only force besides tension is gravity g .
3. The mass is a point particle; air resistance is negligible.

Questions.

- Q1. Over the horizontal.** Derive the minimum v_0 such that at $t = t'$, $\theta > \pi/2$ (the swing has passed the horizontal).
- Q2. Over the top.** Derive the minimum v_0 such that at $t = t'$, $\theta > \pi$ (the swing has passed the vertical).
- Q3. The point of no return.** Find the minimum angle θ^* with the following property: if the swing reaches $\theta = \theta^*$ with the string still taut, then the tension is positive at every angle beyond θ^* . (In other words, θ^* is the point of no return —once passed with the string taut, the swing is guaranteed to remain taut for the rest of the revolution.)